

Vo Hong Quan

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github.com/vohongquann | vohongquann.github.io/Portfolio

Education

Ho Chi Minh City University of Technology and Engineering, Ho Chi Minh City, Vietnam Expected Dec 2026
Bachelor of Robotics and Artificial Intelligence GPA: 3.22/4.0
Relevant coursework: Machine Vision, Artificial Intelligence, Artificial Neural Networks, Practice of Artificial Intelligence

Experience

Robotics Lab, Ho Chi Minh City University of Technology and Engineering Feb 2025 – Present
Research Assistant Ho Chi Minh City, Vietnam

- Train and evaluate reinforcement learning controllers for real-time robotic systems.
- Design reward functions and tune hyperparameters for stable policies.

Projects

Crazyflie-IsaacLab: RL-Tuned PID Gains for a Crazyflie Quadrotor Jan 2026 – Sep 2026
Python, PyTorch, PPO (rsl_rl), NVIDIA Isaac Lab github.com/vohongquann/Crazyflie-IsaacLab

- Built a four-layer flight controller (50–500 Hz) and physics-based Crazyflie model in Isaac Lab 3.0.
- Trained one PPO MLP per layer to tune 9 PID gains, reducing tracking error 23–62% versus PID (attitude: 15.5° to 5.9°).
- Anchored zero network output to the stable PID baseline and sized each MLP for STM32 deployment.
- Measured 2.6–6.3× more outer-loop motor chatter and documented it as the next improvement.
- Wrote simulator-free tests and reproducible reporting scripts.

BalancingRobot-IsaacLab: Self-Balancing Robot with an RL Cascade Jun 2025 – Dec 2025
Python, PPO (rsl_rl), ROS 2, ONNX, LQR, PID github.com/vohongquann/BalancingRobot-IsaacLab

- Trained a PPO velocity policy that maps commands and raw IMU/encoder data to wheel torques.
- Added a frozen position policy; recorded zero falls in 768 randomized episodes and 8 mm median goal error.
- Beat LQR on speed steps: 0.026 m/s error in 0.36 s versus 0.067 m/s in 1.1 s.
- Achieved 0.043 m median figure-eight error with an RL-tuned PID, best among four controllers.
- Deployed ONNX policies in a ROS 2 bridge from sensors to wheel torque.

RotaryPendulum-IsaacLab: Learned Cascade PID for a Rotary Inverted Pendulum Feb 2025 – May 2025
Python, PPO (rsl_rl), NVIDIA Isaac Lab, PhysX, URDF github.com/vohongquann/RotaryPendulum-IsaacLab

- Modeled the QUBE-Servo 2 from CAD with a DC motor and quantized encoders at 500 Hz.
- Trained a PPO policy on 4,096 robots in 64 minutes to tune six cascade-PID gains at 100 Hz.
- Validated on 256 randomized robots: 0.40 s swing-up, 0.23–0.50 s settling and at most 1.4° median overshoot.
- Published weights, a 1080p demo and one-command evaluation scripts.

Skills

Programming: Python, C/C++, MATLAB (Simulink, Simscape, RL Toolbox)
Reinforcement Learning & Control: PPO, SAC, reward design, hyperparameter tuning, cascaded PID, LQR
Robotics & Simulation: NVIDIA Isaac Sim, Isaac Lab, PhysX, ROS 2, URDF, ESP32
Machine Learning & Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO
Tools: Git, GitHub, Linux, Conda, Jupyter/Colab, pytest
Languages: Vietnamese (native), English (TOEIC 715/990)

Certifications

TOEIC Listening & Reading: 715/990, IIG Vietnam 2025